8 **(a)** In the first decade of the 20th century Albert Einstein studied the photoelectric effect. The photoelectric current I for potential difference V between the photocathode and the anode was first measured. The experiment was then repeated using light of the same frequency but of different intensity.

The series of graphs of I for potential difference V produced were similar to those drawn in Fig 8.1.

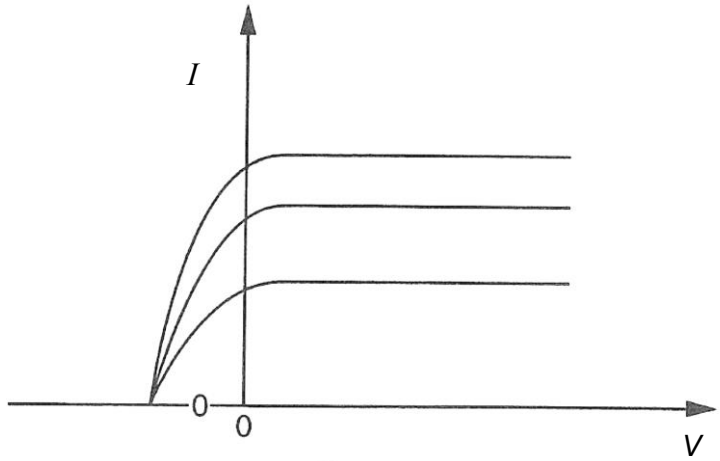

Fig. 8.1

State which feature of this graph could not be explained by Einstein using the wave theory of light.

……

……

…… [2]

(b) Fig. 8.2 shows a photocell using cesium as the photocathode. When the cesium surface is exposed to electromagnetic radiation, photoelectrons are ejected. The anode collects the photoelectrons and the sensitive ammeter indicates the presence of a tiny current.

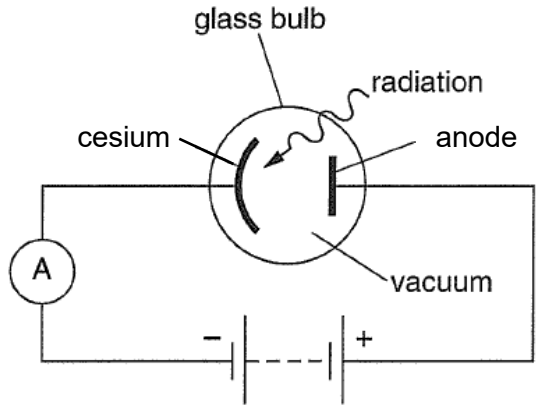

Fig. 8.2

(i) For a certain frequency and intensity of radiation, the ammeter shows a current of 1.2×10^{-7} A. Calculate the number of photoelectrons reaching the collector in 5.0 s.

number of photoelectrons = ……. [2]

(ii) The work function energy of cesium is 2.2 eV and the incident radiation has frequency 7.0×10^{14} Hz. Calculate the maximum kinetic energy of the photoelectrons.

maximum kinetic energy = ……. J [2]

(iii) The intensity of the incident radiation is doubled but the wavelength is kept constant. State and explain the effect this has on each of the following

1. the maximum kinetic energy of the photoelectrons,

…….....

..……

……..

…… [3]

2. the current in the photocell.

…….....

…….....

…… [2]

(iv) Explain why, no current is detected by the ammeter when the cesium photocathode is replaced with zinc.

…….....

…….....

…….. [2]

(c) Einstein's observation introduced the particulate nature of light waves.

(i) Describe an experiment that shows the wave nature of a particle.

…….....

…….

…….....

…… [2]

(ii) Calculate the de Broglie wavelength for a single electron which has a speed of 2.5×10^6 m s^{-1}.

wavelength = ……. m [1]

(iii) State and explain whether electrons with speed of 2.5×10^6 m s^{-1} are suitable to be used to study atomic structure of matter.

…….....

…… [1]

(iv) Neutrons can also be used to investigate the atomic structure of matter. Suggest and explain an advantage and a disadvantage in using neutrons.

advantage ……..

…….....

…….....

disadvantage …....…….....

…….....

…… [3]